

Отчёт по лабораторной работе №5

Эмуляция и измерение потерь пакетов в глобальных сетях

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1 Цель работы

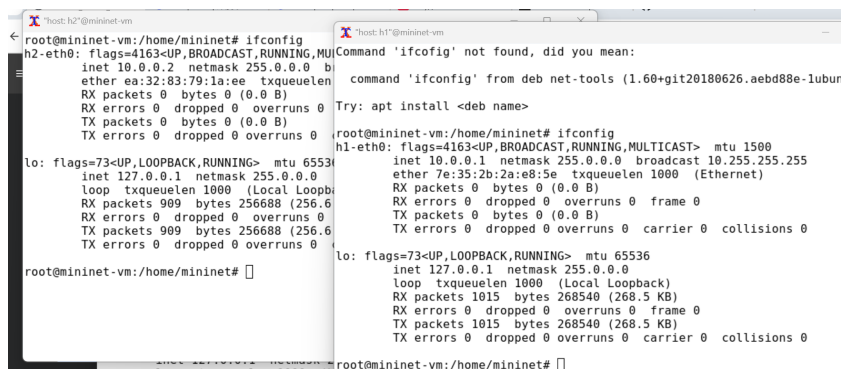
Основной целью работы является получение навыков проведения интерактивных экспериментов в среде Mininet по исследованию параметров сети, связанных с потерей, дублированием, изменением порядка и повреждением пакетов при передаче данных. Эти параметры влияют на производительность протоколов и сетей.

2 Задание

1. Задайте простейшую топологию, состоящую из двух хостов и коммутатора с назначенной по умолчанию mininet сетью 10.0.0.0/8.
2. Проведите интерактивные эксперименты по исследованию параметров сети, связанных с потерей, дублированием, изменением порядка и повреждением пакетов при передаче данных.
3. Реализуйте воспроизводимый эксперимент по добавлению правила отбрасывания пакетов в эмулируемой глобальной сети. На экран выведите сводную информацию о потерянных пакетах.
4. Самостоятельно реализуйте воспроизводимые эксперименты по исследованию параметров сети, связанных с потерей, изменением порядка и повреждением пакетов при передаче данных. На экран выведите сводную информацию о потерянных пакетах.

3 Выполнение лабораторной работы

Задал простейшую топологию, просмотрел ifconfig (рис. 3.1)



```
root@mininet-virtual-machine: /home/mininet# ifconfig
h2-eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.0.2 netmask 255.0.0.0 broadcast 10.255.255.255
    ether ea:32:83:79:1a:ee txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    loop txqueuelen 1000 (Local Loopback)
    RX packets 909 bytes 256688 (256.6 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 909 bytes 256688 (256.6 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

root@mininet-virtual-machine: /home/mininet#

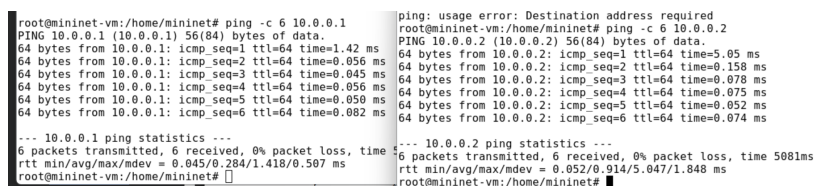
root@mininet-virtual-machine: /home/mininet# ifconfig
h1-eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.0.1 netmask 255.0.0.0 broadcast 10.255.255.255
    ether 7e:35:b2:a8:e8:5e txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    loop txqueuelen 1000 (Local Loopback)
    RX packets 1015 bytes 268540 (268.5 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 1015 bytes 268540 (268.5 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

root@mininet-virtual-machine: /home/mininet#
```

Рис. 3.1: ifconfig

Пропинговал (рис. 3.2)



```
root@mininet-virtual-machine: /home/mininet# ping -c 6 10.0.0.1
PING 10.0.0.1 (10.0.0.1) 56(84) bytes of data:
64 bytes from 10.0.0.1: icmp_seq=1 ttl=64 time=1.42 ms
64 bytes from 10.0.0.1: icmp_seq=2 ttl=64 time=0.056 ms
64 bytes from 10.0.0.1: icmp_seq=3 ttl=64 time=0.045 ms
64 bytes from 10.0.0.1: icmp_seq=4 ttl=64 time=0.056 ms
64 bytes from 10.0.0.1: icmp_seq=5 ttl=64 time=0.050 ms
64 bytes from 10.0.0.1: icmp_seq=6 ttl=64 time=0.082 ms
--- 10.0.0.1 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 500ms
rtt min/avg/max/mdev = 0.045/0.284/1.418/0.507 ms
root@mininet-virtual-machine: /home/mininet#

root@mininet-virtual-machine: /home/mininet# ping -c 6 10.0.0.2
ping: usage error: Destination address required
root@mininet-virtual-machine: /home/mininet# ping -c 6 10.0.0.2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data:
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=5.05 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.158 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.078 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.075 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=0.052 ms
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=0.074 ms
--- 10.0.0.2 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 500ms
rtt min/avg/max/mdev = 0.052/0.914/5.047/1.848 ms
root@mininet-virtual-machine: /home/mininet#
```

Рис. 3.2: Пингование

На h1 добавил 10% потерь пакетов (рис. 3.3)

```

root@mininet-vm:/home/mininet# sudo tc qdisc add dev h1-eth0 root netem loss 10%
root@mininet-vm:/home/mininet# ping -c 100 10.0.0.2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data:
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=0.217 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.084 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.163 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.130 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=0.079 ms
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=0.049 ms
64 bytes from 10.0.0.2: icmp_seq=7 ttl=64 time=0.095 ms
64 bytes from 10.0.0.2: icmp_seq=8 ttl=64 time=0.088 ms
--- 10.0.0.2 ping statistics ---
100 packets transmitted, 90 received, 10% packet loss, time 101383ms
rtt min/avg/max/mdev = 0.039/0.071/0.217/0.026 ms
root@mininet-vm:/home/mininet#

```

Рис. 3.3: На h1 10% потерь пакетов

10% packet loss

На h2 тоже добавил 10% потерь пакетов (рис. 3.4)

```

root@mininet-vm:/home/mininet# sudo tc qdisc add dev h2-eth0 root netem loss 10%
root@mininet-vm:/home/mininet#
64 bytes from 10.0.0.2: icmp_seq=98 ttl=64 time=0.091 ms
64 bytes from 10.0.0.2: icmp_seq=99 ttl=64 time=0.076 ms
64 bytes from 10.0.0.2: icmp_seq=100 ttl=64 time=0.109 ms
--- 10.0.0.2 ping statistics ---
100 packets transmitted, 78 received, 22% packet loss, time 101359ms
rtt min/avg/max/mdev = 0.045/0.104/1.227/0.145 ms
root@mininet-vm:/home/mininet#

```

Рис. 3.4: На h2 тоже 10% потерь пакетов

22% packet loss

Добавил на интерфейсе узла h1 коэффициент потери пакетов 50% (рис. 3.5)


```

rtt min/avg/max/mdev = 0.030/0.062/1.400/1.010 ms
root@mininet-virtual-machine:~# sudo tc qdisc add dev h1-eth0 root netem loss 50
% 50%

64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=2.75 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.618 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.138 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.058 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=0.082 ms
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=0.074 ms
64 bytes from 10.0.0.2: icmp_seq=7 ttl=64 time=0.078 ms
64 bytes from 10.0.0.2: icmp_seq=8 ttl=64 time=0.068 ms
64 bytes from 10.0.0.2: icmp_seq=9 ttl=64 time=0.074 ms
64 bytes from 10.0.0.2: icmp_seq=10 ttl=64 time=0.052 ms
64 bytes from 10.0.0.2: icmp_seq=11 ttl=64 time=0.052 ms
64 bytes from 10.0.0.2: icmp_seq=12 ttl=64 time=0.050 ms
64 bytes from 10.0.0.2: icmp_seq=13 ttl=64 time=0.063 ms
64 bytes from 10.0.0.2: icmp_seq=14 ttl=64 time=0.232 ms
64 bytes from 10.0.0.2: icmp_seq=15 ttl=64 time=0.051 ms
64 bytes from 10.0.0.2: icmp_seq=16 ttl=64 time=0.047 ms
64 bytes from 10.0.0.2: icmp_seq=17 ttl=64 time=0.067 ms
64 bytes from 10.0.0.2: icmp_seq=18 ttl=64 time=0.059 ms
64 bytes from 10.0.0.2: icmp_seq=19 ttl=64 time=0.066 ms
64 bytes from 10.0.0.2: icmp_seq=20 ttl=64 time=0.054 ms
64 bytes from 10.0.0.2: icmp_seq=21 ttl=64 time=0.055 ms
64 bytes from 10.0.0.2: icmp_seq=22 ttl=64 time=0.096 ms
64 bytes from 10.0.0.2: icmp_seq=23 ttl=64 time=0.077 ms
64 bytes from 10.0.0.2: icmp_seq=24 ttl=64 time=0.052 ms

--- 10.0.0.2 ping statistics ---
50 packets transmitted, 23 received, 54% packet loss, time 50154ms
rtt min/avg/max/mdev = 0.047/0.215/2.752/0.553 ms
root@mininet-virtual-machine:~#

```

Рис. 3.5: На интерфейсе узла h1 коэффициент потери пакетов 50%

Добавил на интерфейсе узла h1 0,01% повреждения пакетов (рис. 3.6)

```

root@mininet-virtual-machine:~# sudo tc qdisc add dev h1-eth0 root netem corrupt
0.01%

root@mininet-virtual-machine:~# iperf3 -s
warning: this system does not seem to support IPv6 - trying IPv4
Server listening on 5201

root@mininet-virtual-machine:~# iperf3 -c 10.0.0.2
Connecting to host 10.0.0.2, port 5201
[ 7] local 10.0.0.1 port 58214 connected to 10.0.0.2 port 5201
[ ID] Interval           Transfer     Bitrate        Retr  Cwnd
[ 7] 0.00-1.00 sec    2.27 GBytes  19.5 Gbits/sec    5   690 KBytes
[ 7] 1.00-2.00 sec    2.10 GBytes  18.1 Gbits/sec    6   694 KBytes
[ 7] 2.00-3.00 sec    2.42 GBytes  20.8 Gbits/sec    5   908 KBytes
[ 7] 3.00-4.00 sec    2.35 GBytes  20.2 Gbits/sec    5   947 KBytes
[ 7] 4.00-5.00 sec    2.14 GBytes  18.3 Gbits/sec    4   793 KBytes
[ 7] 5.00-6.00 sec    2.26 GBytes  19.4 Gbits/sec    4   576 KBytes

```

Рис. 3.6: На интерфейсе узла h1 0,01% повреждения пакетов

25% пакетов - сразу, остальные - с задержкой (рис. 3.7)

```

root@mininet-vm:/home/mininet# sudo tc qdisc add dev h1-eth0 root netem delay 1 ☆
0ms reorder 25% 50%
root@mininet-vm:/home/mininet# ping -c 20 10.0.0.2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=1.45 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=11.3 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=11.0 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=12.2 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=10.4 ms
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=11.9 ms
64 bytes from 10.0.0.2: icmp_seq=7 ttl=64 time=10.4 ms
64 bytes from 10.0.0.2: icmp_seq=8 ttl=64 time=10.9 ms
64 bytes from 10.0.0.2: icmp_seq=9 ttl=64 time=10.6 ms
64 bytes from 10.0.0.2: icmp_seq=10 ttl=64 time=11.0 ms
64 bytes from 10.0.0.2: icmp_seq=11 ttl=64 time=10.4 ms
64 bytes from 10.0.0.2: icmp_seq=12 ttl=64 time=10.8 ms
64 bytes from 10.0.0.2: icmp_seq=13 ttl=64 time=11.1 ms
64 bytes from 10.0.0.2: icmp_seq=14 ttl=64 time=10.8 ms
64 bytes from 10.0.0.2: icmp_seq=15 ttl=64 time=10.9 ms
64 bytes from 10.0.0.2: icmp_seq=16 ttl=64 time=10.6 ms
64 bytes from 10.0.0.2: icmp_seq=17 ttl=64 time=11.0 ms
64 bytes from 10.0.0.2: icmp_seq=18 ttl=64 time=10.8 ms
64 bytes from 10.0.0.2: icmp_seq=19 ttl=64 time=10.6 ms
64 bytes from 10.0.0.2: icmp_seq=20 ttl=64 time=10.2 ms

--- 10.0.0.2 ping statistics ---
20 packets transmitted, 20 received, 0% packet loss, time 19047ms
rtt min/avg/max/mdev = 1.445/10.422/12.163/2.112 ms

```

Рис. 3.7: 25% пакетов - сразу, остальные - с задержкой

h1 задал правило с дублированием 50% пакетов (рис. 3.8)

```

root@mininet-vm:/home/mininet# sudo tc qdisc add dev h1-eth0 root netem duplica
te 50%
root@mininet-vm:/home/mininet# ping -c 20 10.0.0.2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=0.172 ms
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=0.178 ms (DUP!)
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.049 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.057 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.058 ms (DUP!)
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.062 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.063 ms (DUP!)
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=0.085 ms
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=0.063 ms
64 bytes from 10.0.0.2: icmp_seq=7 ttl=64 time=0.064 ms
64 bytes from 10.0.0.2: icmp_seq=8 ttl=64 time=0.052 ms
64 bytes from 10.0.0.2: icmp_seq=8 ttl=64 time=0.052 ms (DUP!)
64 bytes from 10.0.0.2: icmp_seq=9 ttl=64 time=0.047 ms
64 bytes from 10.0.0.2: icmp_seq=10 ttl=64 time=0.049 ms
64 bytes from 10.0.0.2: icmp_seq=11 ttl=64 time=0.067 ms
64 bytes from 10.0.0.2: icmp_seq=12 ttl=64 time=0.088 ms
64 bytes from 10.0.0.2: icmp_seq=12 ttl=64 time=0.088 ms (DUP!)
64 bytes from 10.0.0.2: icmp_seq=13 ttl=64 time=0.053 ms
64 bytes from 10.0.0.2: icmp_seq=14 ttl=64 time=0.052 ms
64 bytes from 10.0.0.2: icmp_seq=15 ttl=64 time=0.083 ms
64 bytes from 10.0.0.2: icmp_seq=16 ttl=64 time=0.081 ms
64 bytes from 10.0.0.2: icmp_seq=16 ttl=64 time=0.082 ms (DUP!)
64 bytes from 10.0.0.2: icmp_seq=17 ttl=64 time=0.079 ms
64 bytes from 10.0.0.2: icmp_seq=18 ttl=64 time=0.049 ms
64 bytes from 10.0.0.2: icmp_seq=19 ttl=64 time=0.062 ms
64 bytes from 10.0.0.2: icmp_seq=19 ttl=64 time=0.063 ms (DUP!)
64 bytes from 10.0.0.2: icmp_seq=20 ttl=64 time=0.128 ms

```

Рис. 3.8: Правило с дублированием 50% пакетов

Написал скрипт (рис. 3.9)

```

mininet@mininet-vm: ~/work/lab_netem_ii/simple-drop
GNU nano 4.8 lab_netem_ii.py

info('*** Adding controller\n')
net.addController('c0')

info('*** Adding hosts\n')
h1 = net.addHost('h1', ip='10.0.0.1')
h2 = net.addHost('h2', ip='10.0.0.2')

info('*** Adding switch\n')
s1 = net.addSwitch('s1')

info('*** Creating links\n')
net.addLink(h1, s1)
net.addLink(h2, s1)

info('*** Starting network\n')
net.start()

info('*** Setting packet loss 10% on both interfaces\n')
h1.cmdPrint('tc qdisc add dev h1-eth0 root netem loss 10%')
h2.cmdPrint('tc qdisc add dev h2-eth0 root netem loss 10%')

time.sleep(2) # Wait 2 seconds

info('*** Running ping test\n')
# ...
ping_output = h1.cmd('ping -c 20 ' + h2.IP())
# ...
info('*** Ping results:\n')
print("=" * 50)
print("PING TEST RESULTS:")
print("=" * 50)

```

Рис. 3.9: python скрипт

Создал Makefile (рис. 3.10)

```

GNU nano 4.8 Makefile
all: ping.dat

ping.dat:
    sudo python lab_netem_ii.py
    sudo chown mininet:mininet ping.dat

clean:
    -rm -f *.dat

```

Рис. 3.10: Makefile

Запуск make (рис. 3.11)

```

0:=====
PING TEST RESULTS:
=====
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.604 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.218 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=0.069 ms
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=0.109 ms
64 bytes from 10.0.0.2: icmp_seq=7 ttl=64 time=0.066 ms
64 bytes from 10.0.0.2: icmp_seq=8 ttl=64 time=0.113 ms
64 bytes from 10.0.0.2: icmp_seq=9 ttl=64 time=0.061 ms
64 bytes from 10.0.0.2: icmp_seq=10 ttl=64 time=0.051 ms
64 bytes from 10.0.0.2: icmp_seq=11 ttl=64 time=0.075 ms
64 bytes from 10.0.0.2: icmp_seq=12 ttl=64 time=1.35 ms
64 bytes from 10.0.0.2: icmp_seq=13 ttl=64 time=0.068 ms
64 bytes from 10.0.0.2: icmp_seq=15 ttl=64 time=0.047 ms
64 bytes from 10.0.0.2: icmp_seq=16 ttl=64 time=0.080 ms
64 bytes from 10.0.0.2: icmp_seq=17 ttl=64 time=0.071 ms
64 bytes from 10.0.0.2: icmp_seq=18 ttl=64 time=0.045 ms
64 bytes from 10.0.0.2: icmp_seq=19 ttl=64 time=0.089 ms
64 bytes from 10.0.0.2: icmp_seq=20 ttl=64 time=0.063 ms

--- 10.0.0.2 ping statistics ---
20 packets transmitted, 17 received, 15% packet loss, time 19419ms
rtt min/avg/max/mdev = 0.045/0.187/1.353/0.318 ms

=====
*** Saving filtered data to ping.dat

```

Рис. 3.11: Запуск make

Написал lab_netem_loss.py (рис. 3.12)

```

GNU nano 4.8 lab_netem_loss.py

# Create hosts
h1 = net.addHost('h1', ip='10.0.0.1')
h2 = net.addHost('h2', ip='10.0.0.2')
s1 = net.addSwitch('s1')

# Create links
net.addLink(h1, s1)
net.addLink(h2, s1)

# Start network
net.start()

info('*** Setting 15% packet loss on h1\n')
h1.cmd('tc qdisc add dev h1-eth0 root netem loss 15%')

time.sleep(2)

info('*** Running ping test with packet loss\n')

# Run ping and display full results
ping_output = h1.cmd('ping -c 30 10.0.0.2')

print("=" * 60)
print("PACKET LOSS EXPERIMENT RESULTS:")
print("=" * 60)
print(ping_output)
print("=" * 60)

# Save detailed timing data
h1.cmd('ping -c 30 10.0.0.2 | grep "time=" > ping_loss.log')

# Display packet loss statistics
loss_stats = h1.cmd('ping -c 30 10.0.0.2 | grep "packet loss"')

^G Get Help ^O Write Out ^W Where Is ^K Cut Text ^J Justify ^C Cur Pos
^X Exit ^R Read File ^\ Replace ^U Paste Text ^T To Spell ^_ Go To Lin

```

Рис. 3.12: lab_netem_loss.py

Запуск lab_netem_loss.py (рис. 3.13)

```
*** Setting 15% packet loss on h1
*** Running ping test with packet loss
=====
PACKET LOSS EXPERIMENT RESULTS:
=====
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=1.75 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.277 ms
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=0.049 ms
64 bytes from 10.0.0.2: icmp_seq=7 ttl=64 time=0.060 ms
64 bytes from 10.0.0.2: icmp_seq=9 ttl=64 time=0.059 ms
64 bytes from 10.0.0.2: icmp_seq=10 ttl=64 time=0.063 ms
64 bytes from 10.0.0.2: icmp_seq=11 ttl=64 time=0.063 ms
64 bytes from 10.0.0.2: icmp_seq=12 ttl=64 time=0.051 ms
64 bytes from 10.0.0.2: icmp_seq=13 ttl=64 time=0.092 ms
64 bytes from 10.0.0.2: icmp_seq=14 ttl=64 time=0.063 ms
64 bytes from 10.0.0.2: icmp_seq=15 ttl=64 time=0.056 ms
64 bytes from 10.0.0.2: icmp_seq=16 ttl=64 time=0.081 ms
64 bytes from 10.0.0.2: icmp_seq=17 ttl=64 time=0.105 ms
64 bytes from 10.0.0.2: icmp_seq=18 ttl=64 time=0.060 ms
64 bytes from 10.0.0.2: icmp_seq=19 ttl=64 time=0.066 ms
64 bytes from 10.0.0.2: icmp_seq=20 ttl=64 time=0.050 ms
64 bytes from 10.0.0.2: icmp_seq=21 ttl=64 time=0.086 ms
64 bytes from 10.0.0.2: icmp_seq=22 ttl=64 time=0.059 ms
64 bytes from 10.0.0.2: icmp_seq=23 ttl=64 time=0.059 ms
64 bytes from 10.0.0.2: icmp_seq=24 ttl=64 time=0.074 ms
64 bytes from 10.0.0.2: icmp_seq=25 ttl=64 time=0.061 ms
64 bytes from 10.0.0.2: icmp_seq=26 ttl=64 time=0.081 ms
64 bytes from 10.0.0.2: icmp_seq=28 ttl=64 time=0.078 ms
64 bytes from 10.0.0.2: icmp_seq=29 ttl=64 time=0.052 ms
64 bytes from 10.0.0.2: icmp_seq=30 ttl=64 time=0.057 ms

--- 10.0.0.2 ping statistics ---
30 packets transmitted, 25 received, 16.6667% packet loss, time 29639ms
rtt min/avg/max/mdev = 0.049/0.141/1.747/0.330 ms
```

Рис. 3.13: Запуск lab_netem_loss.py

Написал lab_netem_reorder.py (рис. 3.14)

```
GNU nano 4.8 lab_netem_reorder.py
#!/usr/bin/env python

"""
Experiment with packet reordering
"""

from mininet.net import Mininet
from mininet.node import Controller
from mininet.log import setLogLevel, info
import time

def reorder_experiment():
    "Experiment with packet reordering"

    net = Mininet(controller=Controller, waitConnected=True)
    net.addController('c0')

    # Create hosts
    h1 = net.addHost('h1', ip='10.0.0.1')
    h2 = net.addHost('h2', ip='10.0.0.2')
    s1 = net.addSwitch('s1')

    # Create links
    net.addLink(h1, s1)
    net.addLink(h2, s1)

    # Start network
    net.start()

    info('*** Setting packet reordering: 25% packets reordered with 50ms delay\n')
    h1.cmd('tc qdisc add dev h1-eth0 root netem delay 50ms reorder 25%')

    time.sleep(2)

[ Read 67 lines ]
^G Get Help ^O Write Out ^W Where Is ^K Cut Text ^J Justify ^G Cur Pos
^X Exit ^R Read File ^\ Replace ^U Paste Text ^T To Spell ^_ Go To L
```

Рис. 3.14: lab_netem_reorder.py

Запуск lab_netem_reorder.py (рис. 3.15)

```

s1
*** Setting packet reordering: 25% packets reordered with 50ms delay
*** Running ping test with packet reordering
=====
PACKET REORDERING EXPERIMENT RESULTS:
=====
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=51.8 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=50.6 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=50.6 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=50.3 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=0.077 ms
64 bytes from 10.0.0.2: icmp_seq=6 ttl=64 time=0.065 ms
64 bytes from 10.0.0.2: icmp_seq=7 ttl=64 time=50.3 ms
64 bytes from 10.0.0.2: icmp_seq=8 ttl=64 time=50.5 ms
64 bytes from 10.0.0.2: icmp_seq=9 ttl=64 time=50.3 ms
64 bytes from 10.0.0.2: icmp_seq=10 ttl=64 time=0.097 ms
64 bytes from 10.0.0.2: icmp_seq=11 ttl=64 time=50.3 ms
64 bytes from 10.0.0.2: icmp_seq=12 ttl=64 time=50.4 ms
64 bytes from 10.0.0.2: icmp_seq=13 ttl=64 time=50.9 ms
64 bytes from 10.0.0.2: icmp_seq=14 ttl=64 time=50.7 ms
64 bytes from 10.0.0.2: icmp_seq=15 ttl=64 time=50.2 ms
64 bytes from 10.0.0.2: icmp_seq=16 ttl=64 time=50.5 ms
64 bytes from 10.0.0.2: icmp_seq=17 ttl=64 time=0.075 ms
64 bytes from 10.0.0.2: icmp_seq=18 ttl=64 time=51.0 ms
64 bytes from 10.0.0.2: icmp_seq=19 ttl=64 time=51.6 ms
64 bytes from 10.0.0.2: icmp_seq=20 ttl=64 time=50.5 ms

--- 10.0.0.2 ping statistics ---
20 packets transmitted, 20 received, 0% packet loss, time 19127ms
rtt min/avg/max/mdev = 0.065/40.545/51.817/20.237 ms

```

Рис. 3.15: Запуск lab_netem_reorder.py

Написал lab_netem_corruption.py (рис. 3.16)

```

#!/usr/bin/env python
"""
Experiment with packet corruption
"""
from mininet.net import Mininet
from mininet.node import Controller
from mininet.log import setLogLevel, info
import time

def corrupt_experiment():
    "Experiment with packet corruption"

    net = Mininet(controller=Controller, waitConnected=True)
    net.addController('c0')

    # Create hosts
    h1 = net.addHost('h1', ip='10.0.0.1')
    h2 = net.addHost('h2', ip='10.0.0.2')
    s1 = net.addSwitch('s1')

    # Create links
    net.addLink(h1, s1)
    net.addLink(h2, s1)

    # Start network
    net.start()

    info('*** Setting 1% packet corruption on h1\n')
    h1.cmd('tc qdisc add dev h1-eth0 root netem corrupt 1%')

    time.sleep(2)

```

Рис. 3.16: Написал lab_netem_corruption.py

Запуск lab_netem_corruption.py (рис. 3.17)

```

IPERF3 TEST (10 seconds):
Connecting to host 10.0.0.2, port 5201
[ 5] local 10.0.0.1 port 58236 connected to 10.0.0.2 port 5201
[ ID] Interval      Transfer    Bitrate      Retr  Cwnd
[ 5]  0.00-1.00    sec  1.35 GBytes  11.6 Gbits/sec  548   84.8 KBytes
[ 5]  1.00-2.00    sec  1.06 GBytes  9.13 Gbits/sec  406   161 KBytes
[ 5]  2.00-3.00    sec  1.42 GBytes  12.2 Gbits/sec  465   144 KBytes
[ 5]  3.00-4.00    sec  1.32 GBytes  11.3 Gbits/sec  439   168 KBytes
[ 5]  4.00-5.00    sec  1.18 GBytes  10.2 Gbits/sec  392   105 KBytes
[ 5]  5.00-6.00    sec  1.33 GBytes  11.4 Gbits/sec  432   59.4 KBytes
[ 5]  6.00-7.00    sec  1.13 GBytes  9.68 Gbits/sec  369   247 KBytes
[ 5]  7.00-8.00    sec  1.07 GBytes  9.23 Gbits/sec  370   720 KBytes
[ 5]  8.00-9.00    sec  1.40 GBytes  12.0 Gbits/sec  450   321 KBytes
[ 5]  9.00-10.00   sec  1.11 GBytes  9.51 Gbits/sec  362   202 KBytes
- - - - -
[ ID] Interval      Transfer    Bitrate      Retr
[ 5]  0.00-10.00   sec  12.4 GBytes  10.6 Gbits/sec  4233
[ 5]  0.00-10.00   sec  12.4 GBytes  10.6 Gbits/sec
sender
receiver
iperf Done.

```

Рис. 3.17: Запуск lab_netem_corruption.py

4 Выводы

Получил навыки проведения интерактивных экспериментов в среде Mininet по исследованию параметров сети, связанных с потерей, дублированием, изменением порядка и повреждением пакетов при передаче данных.

5 Список литературы

1. Официальный сайт и репозиторий Mininet

- Mininet Official Website. – URL: <http://mininet.org/>
- Mininet GitHub Repository. – URL: <https://github.com/mininet/mininet>

2. Документация по системам виртуализации и графическому интерфейсу

- VirtualBox Official Manual. – URL: <https://www.virtualbox.org/manual/UserManual.html>
- X Window System Protocol Documentation. – URL: <https://www.x.org/releases/current/doc/xproto/x11protocol.html>

3. Вспомогательные ресурсы

- VcXsrv Windows X Server. – URL: <https://sourceforge.net/projects/vcxsrv/>
- Xming X Server. – URL: <http://www.straightrunning.com/XmingNotes/>